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*Where Minds Pollinate*

**Computer Science**  
**Class 10**



**FBISE**

**Chapter 2**

**Exercise Solution**

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## Computer Science 10<sup>th</sup>

### Chapter no 2

#### Exercise ( Q/A)

**Q2.1: Consider 3 boys and 3 girls want to team up as pair for the 14th August related performance at the school. Find the total no. of ways these pairs could be formed. Counting Principle Problems.**

Number of Boys = 3

Number of Girls = 3

Total number of ways to pair = (number of boys) \* (number of girls) =  $3 * 3 = 9$

Let A,B,C represents boys and X,Y,Z represents Girls then possible pairs are

(A,X),(A,Y),(A,Z),(B,X),(B,Y),(B,Z),(C,X),(C,Y),(C,Z)

**Q2.2: Suppose you have selected black pant to attend school function and have three options to choose. Find the total no. of ways you can dress up using Counting Principle Problems.**

Total number of ways to dress up =  $1 * 3 = 3$

**Q2.3: You are sitting at a study table doing homework. The penholder on the table contains pens, 1 black pen, and 2 lead pencils. How many options do you have for writing?**

You have 4 options for writing

1. **Black pen** (1 option)
2. **Lead pencil 1** (1 option)
3. **Lead pencil 2** (1 option)
4. **Writing with other pens** (Multiple options, as the prompt states the holder contains "pens" in plural plus the specific 1 black pen and 2 pencils)

**Q2.4: For a race on sports day activities, a group of 20 students are given a race lane at random. There are six students selected for the first race. How many different orders of the six lane students may be chosen?**

There are **27,907,200 different orders** possible for the six lane students.

This is a **permutation** problem because the specific order of the lanes matters.

**Total students (n): 20**

**Lanes to fill (r): 6**

**Permutation:  $p(20,6) = 20!/(20-6)! = 27,907,200$**

**Q3.1: Ejaz, Javeria, Liaqat, and Zainab are standing in a line. Here are the clues to determine positions:**

- a) Javeria is standing directly behind Ejaz.**
- b) Zainab is not next to Liaqat.**
- c) Liaqat is not at the end of the line.**
- d) Zainab is standing in the first position.**

**In which order are they standing?**

The correct standing order from first to last is **Zainab, Ejaz, Liaqat, and Javeria.**

**Position 1:** Zainab is explicitly placed here by clue (d)

**Position 3:** Liaqat must take this spot. He cannot be at the end (position 4) due to clue (c), and he cannot be in position 2 because clue (b) states he isn't next to Zainab.

**Positions 2 and 4:** Ejaz and Javeria fill the remaining gaps. Since clue (a) specifies Javeria is directly behind Ejaz, Ejaz takes position 2 and Javeria takes position 4

**Final Order:**

**1: Zainab**

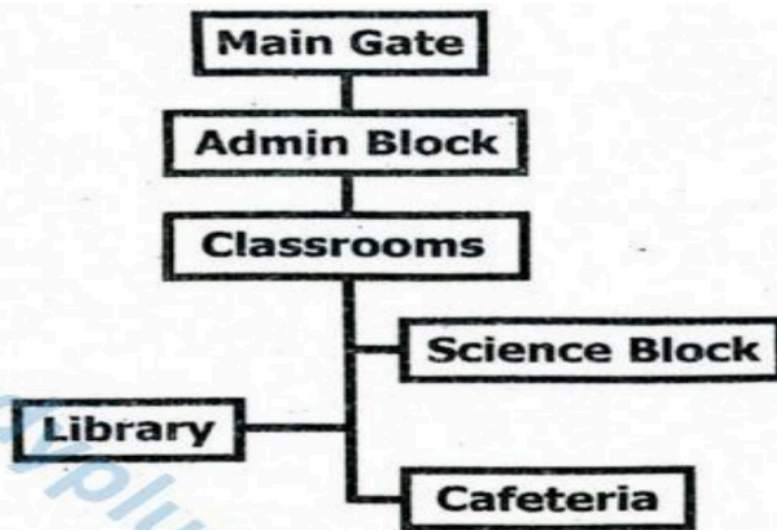
**2: Ejaz**

**3: Liaqat**

**4: Javeria**

**Q3.2: Provide students with a detailed map of their school. Ask them to create an abstract view highlighting only the main buildings and pathways.**

**Abstract view highlighting only the main buildings and pathways :**



**Q3.3: Give students a problem that can be solved with a simple program. Encourage them to write functions to handle different parts of the problem, abstracting complex operations into manageable pieces.**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Basic Calculator</title>
</head>
<body>
  <h2> Simple Calculator</h2>
  <input type="number" id="num1" placeholder="First Number">
  <input type="number" id="num2" placeholder="Second Number">
  <br><br>
  <button onclick="handleCalculate('add')">Add (+)</button>
  <button onclick="handleCalculate('subtract')">Subtract (-)</button>
  <button onclick="handleCalculate('multiply')">Multiply (×)</button>
  <button onclick="handleCalculate('divide')">Divide (÷)</button>
```

```
<h3>Result: <span id="output">0</span></h3>
<script>
// --- Core Math Functions (Abstracted Operations) ---
function add(a, b) { return a + b; }
function subtract(a, b) { return a - b; }
function multiply(a, b) { return a * b; }
function divide(a, b) {
  if (b === 0) return "Error (Cannot divide by 0)";
  return a / b;
}

// --- Main Controller Function ---
function handleCalculate(operation) {
  let n1 = parseFloat(document.getElementById("num1").value) || 0;
  let n2 = parseFloat(document.getElementById("num2").value) || 0;
  let result = 0;

  if (operation === 'add') result = add(n1, n2);
  else if (operation === 'subtract') result = subtract(n1, n2);
  else if (operation === 'multiply') result = multiply(n1, n2);
  else if (operation === 'divide') result = divide(n1, n2);

  document.getElementById("output").innerText = result;
}

</script>
</body>
</html>
```

**Q3.4: Given the following web form inputs, identify any syntax errors and correct them.**

**Form Inputs:**

→ **Name:** Amjad Ali

→ **Email:** amjadali@gmail

→ **Date:** 01/01/2024

The entry **Email: amjadali@gmail** has a syntax error because it is missing a top-level domain (such as .com).

The other inputs are syntactically valid.

**Q3.5: You have developed a web application that takes the budget head and provides total. Analyze the following budget and identify any logical errors in the total amount.**

**Budget Example:**

→ **Rent:** \$1000

→ **Utilities:** \$150

→ **Groceries:** \$200

→ **Total:** \$1000 + \$150 + \$200 = \$1500

The logical error is in the **addition calculation**, which results in an incorrect total.

**The Error**

**The Formula:** \$1000 + \$150 + \$200 equals **\$1350**, not \$1500.

**Correct Budget**

Rent: \$1000

Utilities: \$150

Groceries: \$200

Total: **\$1350**